



Experimental and kinetic modeling studies of 2-ethylfuran pyrolysis at low and atmospheric pressures

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ARTICLE INFO

Article history:

Received 14 August 2020

Revised 17 December 2020

Accepted 17 December 2020

Available online 2 January 2021

Keywords:

Ethylfuran

Methylfuran

Flow reactor

Pyrolysis

SVUV-PIMS

Kinetic model

ABSTRACT

The pyrolysis of 2-ethylfuran (EF2) in a flow reactor was studied by using synchrotron vacuum ultra-violet photoionization mass spectrometry at 846 – 1319 K and at 30 and 760 Torr. Over 20 pyrolysis products were detected and measured. A detailed kinetic model was constructed by extending the reported comprehensive 2,5-dimethylfuran model to describe the pyrolysis of EF2 and validated against the current experimental results, including methyl, 2-furylmethyl, propargyl, acetylene, ethylene, propene, ketene, 2-vinylfuran, furan/vinylketene and benzene, etc. Rate of production analysis and sensitivity analysis indicate that the main reaction pathways of EF2 pyrolysis are H-abstraction reactions forming 1-(2-furyl)ethyl and 2-(2-furyl)ethyl, unimolecular decomposition reactions forming 2-furylmethyl, and H-addition reactions forming furan and vinylketene. The corresponding products, 2-vinylfuran, 2-furylmethyl and vinylketene, etc., are indicated to be the most key products in EF2 pyrolysis. Comparisons of the formation of aromatic hydrocarbons in the pyrolysis of EF2, 2-methylfuran and 2,5-dimethylfuran show that EF2 had a lower tendency to produce aromatic soot precursors.

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1. Introduction

Energy shortage and environmental pollution are two major problems confronting the globe. Alternative energy sources, especially biomass fuels, may reduce the use of fossil energy and hence greenhouse gas emissions. With the development in the transformations of furan series biofuels from biomasses [1–3], furan series biofuels, such as 2-methylfuran (MF2), 2,5-dimethylfuran (DMF25) and 2-ethylfuran (EF2), etc., have drawn much attention. In the early days, Grela et al. [4] conducted pyrolysis experiments on DMF25, MF2 and furan in a heatable molecular flow reactor. They proposed that DMF25 could decompose into benzene + water ($C_6H_6 + H_2O$) and $C_5H_8 + \text{carbon monoxide (CO)}$ in two ways. Lifshitz et al. [5] conducted pyrolysis experiments of furan, MF2 and DMF25 in a single pulse shock tube, using a gas chromatograph to analyze the final products, as well as constructed a preliminary kinetic model. Recently, Cheng et al. [6–8] systematically

researched the pyrolysis of DMF25, MF2, furan in a flow reactor at different pressures. They studied the pyrolysis of DMF25 under 30, 150 and 760 Torr and developed a kinetic model to simulate their experimental results [6]. The primary reaction pathways of DMF25 decomposition and aromatic hydrocarbon formation were described. Cheng et al. [7,8] also studied MF2 pyrolysis at 900 – 1530 K and at 30 and 760 Torr, and furan pyrolysis at 30 Torr. Furthermore, the generation pathways of aromatics in furan, MF2 and DMF25 pyrolysis were also compared in their works. Somers et al. developed a comprehensive DMF25 model [9] based on their previous MF2 model [10] and the DMF25 model of Sirjean et al. [11]. It was verified by most of the previous experimental results. Alexandrino et al. [12] carried out experimental and kinetic modeling researches on DMF25. The effects of temperatures, equivalence ratios, concentration of DMF25 and pressures on its conversion rate were analyzed. Weber et al. [13] detected hydrogen atom (H atom) by atomic resonance absorption spectrometry to study the detailed kinetic processes in the pyrolysis of furan, MF2 and DMF25. Most recently, Su et al. [14] researched the pyrolysis of 3-methylfuran under 30 Torr and 760 Torr and developed a kinetic model to simulate their experimental results, which further developed the kinetic mechanism of furan series fuels.

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As an isomer of DMF25, EF2 has the potential of being a new alternative biofuel. Following the chemical route in the production of DMF25 [1,2], EF2 might be produced in a modified method. EF2 is also an important intermediate in the flames and pyrolysis of MF2 [15,16], as well as in the oxidation of DMF25 [9,17,18]. Since the primary products of MF2 consumption, 2-furylmethyl (MF22J), is a resonantly-stabilized radical, combination of MF22J and methyl (CH_3) to form EF2 is inevitable. This reaction should also contribute to the consumption of alkylfurans, which may produce MF22J readily [15–20]. By this sense, studies on EF2 combustion and pyrolysis are pertinent. Bierbach et al. [21] measured the rate coefficient of hydroxyl radical (OH) with furan, MF2, EF2 and DMF25 in synthetic air at 300 ± 2 K and 760 Torr. Regarding the reaction mechanism of OH with EF2. Zhang et al. [22] constructed the overall profile of the double potential energy surface for $\text{OH} + \text{EF2}$ system at G3MP2 and G3MP2B3 levels. The results showed that the reaction of $\text{OH} + \text{EF2}$ was mainly the addition-elimination mechanism. Smith et al. [23] measured the absolute photoionization cross sections (PICS) of EF2 for the first time. El-deeb and Akih-Kumgeh [24] measured the ignition behaviors of EF2 and its isomer DMF25 in a shock tube at 1009 – 1392 K, revealing the higher reactivity of EF2. Xu et al. [25] studied the spontaneous combustion of EF2 at 766 – 1013 K in a rapid compression machine to measure the ignition delays of different fuels and to compare the reactivities of EF2 with MF2 and DMF25 at different equivalence ratios. The EF2 sub-mechanism also has been further developed based on Somers' model [9].

However, experimental and theoretical researches on EF2 are still scarce currently, especially the understanding of the thermal decomposition mechanism, which is of great significance for studying the exhaust emission and pollutant formation processes. In the present work, the pyrolysis of EF2 was studied experimentally in a flow reactor at 30 Torr and 760 Torr, synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) was used to detected and measured the pyrolysis intermediate species and products. A detailed pyrolysis sub-mechanism of EF2 was updated based on Somers' mechanism [9] and Cheng's mechanism [7]. Rate of production (ROP) analysis and sensitivity analysis were also performed to study the decomposition network of EF2.

2. Experimental method

The experiments were carried out at the National Synchrotron Radiation Laboratory in Hefei, China. Detailed descriptions of the pyrolysis devices were reported elsewhere [26–33]. Figure 1 shows the schematic diagram of the pyrolysis apparatus. In short, the pyrolysis device consisted of three parts, a pyrolysis chamber with a heating furnace (Region I), a differentially pumped chamber with a molecular-beam sampling system (Region II) and a reflective time-of-flight mass spectrometer (Region III). The liquid fuel (EF2, boiling point 365.5 K@1 atm) was filled in the injector (Hamilton81656, 50 mL) and pumped by the high-pressure infusion pump (FL2200, China Fuli Co., Ltd.) to be vaporized and mixed with argon (argon 1, Ar 1) at 120 °C (393 K) in the evaporator. The pre-heated mixture was further diluted with another Ar (Ar 2) flow and krypton (Kr) flow before being transferred to and being pyrolyzed in the α -alumina tube with an internal diameter of 7.0 mm and a length of 220 mm mounted in the furnace in Region I. The flows of Kr and Ar were controlled by MKS mass flow controllers (GM-50A, MKS Instruments, Inc.). The pressure in the pyrolysis chamber was controlled by an MKS throttle valve (253B-20-40-2, MKS Instruments, Inc.) and was kept at 30 or 760 Torr, corresponding to 3.99×10^3 and 1.01×10^5 Pa, respectively. The residence time was estimated to be 20 and 500 ms at 30 and 760 Torr, respectively. Details of the flow rates of fuel and gases in the experiments are shown in Table 1. The pyrolysis species were sampled at 10 mm

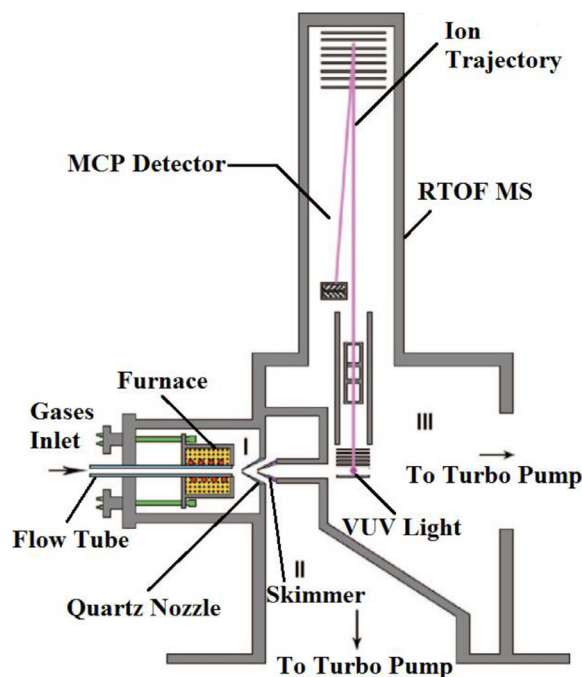


Fig. 1. Schematic diagram of the pyrolysis apparatus with molecular-beam sampling mass spectrometer. Regions I, II and III are pyrolysis chamber, differentially pumped chamber and photoionization chamber, respectively. The synchrotron vacuum ultraviolet (VUV) light crosses vertically with the molecular beam and the ion trajectory.

Table 1

Initial flow rates of 2-ethylfuran (EF2), argon (Ar) and krypton (Kr) in the experiments. The flow rates were measured at 298 K. Units of the flow rates are in $\text{mL}\cdot\text{min}^{-1}$ or standard cubic centimeter per minute (sccm) for liquid and gases, respectively.

Pressure	EF2 (liquid)	EF2 (equivalent gas)	Ar 1	Ar 2	Kr
30 Torr	0.047	10	300	680	10
760 Torr	0.047	10	300	680	10

downstream the exit of the furnace by the quartz nozzle and collimated by the skimmer in Region II to form a molecular beam and entered into the photoionization chamber, i.e., Region III, to be ionized by the synchrotron vacuum ultraviolet. The ions produced therein were detected and measured by the mass spectrometer. EF2 with a purity of 99% was purchased from Meryer (Shanghai) Chemical Technology Co., Ltd., China. Ar and Kr with a purity of 99.999% were purchased from Nanjing Special gases Factory Ltd., China.

Mole fraction convolution, temperature distribution measurements and pressure calculations along the centerline of the reactor were described elsewhere [34–36]. The centerline temperature distributions, i.e., temperature profiles, the temperature distribution along the flow reactor were measured by an S-type thermocouple at different temperatures. The uncertainties of temperature were estimated to be ± 30 K. Each temperature profile was named after its maximum temperature value (T_{max}) and was used as the input temperature profile in EF2 pyrolysis simulations. Thus, in analyzing the pyrolysis at T_{max} , it implies that the pyrolysis ran a temperature history of the experimentally measured values under the nominal name of T_{max} . The temperature profiles are provided in the Supplementary Materials. The mass spectra could be recorded at different photon energies to generate the photoionization efficiency (PIE) spectra to identify the pyrolysis intermediates and products. In this work, PIEs were recorded from 7.8 – 11.5 eV, at 1173 and 973 K at 30 and 760 Torr, respectively. The mass spectra could also

be recorded at different photon energies to record the signal intensities and hence to measure the mole fractions of the pyrolysis intermediates and products at various pyrolysis temperatures. In the present work, the mass spectrometer was measured in a temperature range of 846 – 1319 K and at selected photon energies of 8.50, 9.00, 9.50, 10.00, 10.50, 11.00, 13.00 and 14.50 eV to obtain photoionization of the corresponding pyrolysis intermediates and products. In addition, an ionization energy strategy of 15.300 – 15.400 eV with a step width of 0.002 eV was adopted to measure the mole fraction of species with high ionization energies at both pressures. PICSS data of most pyrolysis products are available in the online database [37]. PICSSs of the species reported in this work are provided in the *Supplementary Materials*. Uncertainties of the experimental mole fractions were estimated within 25% for species with known PICSSs and a factor of 2 for those with estimated PICSSs.

3. Theoretical calculation and kinetic modeling

3.1. Theoretical calculation

The Gaussian 09 program [38] was used to conduct quantum chemistry calculations in the present work. Most of the vibration modes were treated as harmonic oscillators. Based on the code MESS [39], the time-dependent RRKM solution was used to calculate the rate coefficients of these selected reactions. Ar was used as the buffer gas, with the L-J parameters of $\sigma = 3.47$ Å and $\varepsilon = 79.2$ cm⁻¹ recommended by Hippler et al. [40]. The energy transfer was assigned a temperature-dependent form, i.e., $\langle \Delta E_{\text{down}} \rangle = 200(T/300)^{0.85}$ cm⁻¹ and the master equation calculations were carried out at temperatures from 100 to 1500 K and pressures from 0.01 to 100 atm. For barrierless dissociation reactions, the rate constants were calculated by using microcanonical variational state theory. CBS-QB3 method was used to calculate frequency and to optimize the structures. With a relaxed scan, each structure increases by 0.1 Å from 1.5 to 4 Å along the minimum energy pathway. The pertinent potential energy surfaces are provided in Fig. S1 in the *Supplementary Materials*.

3.2. Kinetic modeling

The detailed kinetic model of EF2 consisting of 659 species and 3147 reactions was developed by extending the DMF25 [9] and MF2 [7] models to describe the pyrolysis of EF2. In the base mechanisms, the sub-mechanism for C0–C4 species was taken from previously reported mechanisms [41–44]. The DMF25 sub-mechanism was taken from Somers et al. [9] and Sirjean et al. [11]. The MF2 model was taken from Somers et al. and Cheng et al. [7,10,20,45]. The furan model was taken from Tian et al. [46]. In the current mechanism, the aromatics sub-mechanism was mainly based on the recently reported comprehensive toluene models [47,48], ethylbenzene model [49] and tetralin model [35]. Some vital reactions in the sub-mechanism of EF2 are listed in Table 2. According to the previous work of Somers et al. [9], there are four main types of reactions that contributed to the consumption of EF2, namely, the unimolecular decomposition reactions (R1–R7), H-abstraction reactions (R8–R13), H-transfer reactions (R14–R15) and H-addition reactions (R16–R22). Parameters of some key reactions were calculated, including R1–R2, R14–R15 and R23–R26. EF2 sub-mechanism was proposed by analogy to that of MF2 [9], including R5–R7, R16–R19, R27 and R28. Table S1 in the *Supplementary Materials* lists the structures of selected species in the present model. The EF2 sub-mechanism is introduced in detail as follows.

3.2.1. Unimolecular decomposition reactions of EF2

The unimolecular decomposition reactions of EF2 are mainly those by breaking the C–CH₃ bond in the ethyl (C₂H₅) group

(R1) and C(2)–CH₂ bond (R2) to produce 2-furylmethyl + methyl (MF22J + CH₃) and 2-furyl (F2J) + C₂H₅, respectively. The unimolecular C–H bond dissociation reactions of EF2 pyrolysis (R3–R7) can produce five radicals, i.e., 1-(2-furyl)ethyl (E2F2J-A), 2-(2-furyl)ethyl (E2F2J-P), 3-(2-ethyl)furyl (EF23J), 4-(2-ethyl)furyl (EF24J) and 5-(2-ethyl)furyl (EF25J), while the rate constants of R1–R7 were taken from the models of ethylbenzene [50] and MF2 [9] based on similar structures, respectively.

3.2.2. H-abstraction reactions of EF2

H-abstractions are a group of reactions that contribute significantly to EF2 consumption. There are two H-abstraction sites on the side chain of EF2, producing E2F2J-A and E2F2J-P, respectively. Reactions R8–R13 in Table 2 are H-abstraction reactions by H atom, CH₃, and propargyl (C₃H₃), with rate constants coming from the reactions of DMF25 [9], toluene [51] and ethylbenzene [50] based on similar structures.

3.2.3. H-transfer reactions of EF2

Decomposition of carbene intermediate caused by H-transfer reactions was known to be the most important pathways in the pyrolysis of furan [55], MF2 [7], and DMF25 [6]. EF2 has a similar structure to MF2, with the main difference between EF2 and MF2 lying in the different side chain groups (C₂H₅ and CH₃). Somers et al. [9] determined the carbonic rearrangement of EF2 via 4,5-H-transfer to produce hexa-4,5-dien-3-one (CH₃CH₂COCH=C=CH₂, H45DE3O, R14) and 3,2-H-transfer to produce hexa-2,3-dien-1-one (OCHCH=C=CHCH₂CH₃, H23DE1O, R15). The rate constants of H-transfer reactions were obtained by analogy to DMF25 [11] in Somers' model [9]. The pressure-dependent rate constants of these two reactions were updated from the present theoretical work.

3.2.4. H-addition reactions of EF2

H-addition on EF2 (R16–R22) could form four radicals, 2,3-dihydro-2-ethylfuran-3-yl (EF22H23J), 3,3-dihydro-2-ethylfuran-2-yl (EF23H22J), 4,4-dihydro-2-ethylfuran-5-yl (EF24H25J) and 5,5-dihydro-2-ethylfuran-4-yl (EF25H24J). Since the positions connecting the oxygen atom are favorable, i.e., C(2) and C(5), pathways forming EF23H22J and EF24H25J are negligible compared with those forming EF22H23J and EF25H24J. The latter two radicals played key roles in EF2 pyrolysis. Furan, 1-penten-1-yl (C₅H₉1-1) and vinylketene (CH₂CHCHCO) were the products of H-addition on EF2 to C(2) and C(5), respectively, as shown in Table 2 (R20–R22). The rate constants of H-addition to C(2) and C(5) were referred to similar reactions of ethylbenzene model [52], DMF25 models of Somers et al. [9] and Sirjean et al. [53].

3.2.5. Decomposition reactions of the primary radicals

The primary radicals formed by the unimolecular decomposition reactions, the dissociation reactions of C–H bonds, the H-abstraction reactions and the H-addition reactions may decompose by β -scissions. Reactions R23–R28 in Table 2 are the six significant reactions of those primary radicals. The rate constants of R23–R26 were updated from the present theoretical calculation work. R27 and R28 were referred to similar reactions of MF2 [9]. In the current model, the majority of the thermodynamic data were derived from previous DMF25 models [9,11,53,56], furan model [46], *n*-heptane model [57] and ethylbenzene models [50,58]. The THERM software [59] was used to calculate the thermodynamic data of the newly added species in the current model. The GA values were taken from Bruke et al. [60] and Zhang et al. [57] in this work to calculate the thermodynamic data. No modification of the previous thermodynamic data was made in the current mechanism. The methodology in obtaining these thermodynamic data are the same with the previous works [61,62], thus the newly added data should

Table 2Rate constants for selected reactions in the pyrolysis sub-mechanism of EF2, $k = AT^n \exp(-E_a/RT)$. Units are in cm³, s and cal/mol.

Selected Reactions	A	n	E _a	Pressure (Torr)	Reference
Unimolecular decomposition reactions					
R1 EF2 = MF2J + CH ₃	1.02E+11	−0.01	4.89E+04	7.6	Present work
	1.24E+11	0.23	5.26E+04	30	
	1.26E+11	0.40	5.52E+04	76	
	1.94E+11	0.76	6.19E+04	760	
	1.38E+11	1.09	6.71E+04	7600	
R2 EF2 = F2J + C ₂ H ₅	9.74E+06	2.09	8.90E+04	7.6	Present work
	1.09E+07	2.26	9.23E+04	30	
	1.34E+07	2.33	9.42E+04	76	
	1.28E+07	2.43	9.59E+04	760	
	3.68E+07	2.32	9.65E+04	7600	
R3 EF2 = E2F2J-A + H	3.13E+97	−24.24	1.34E+05	30	[50] ^a
	1.25E+82	−19.60	1.27E+05	150	
	3.95E+60	−13.30	1.15E+05	760	
	6.12E+16	−0.60	8.94E+04	76,000	
R4 EF2 = E2F2J-P + H	1.28E+117	−29.95	1.57E+05	30	[50] ^b
	2.18E+101	−25.07	1.52E+05	150	
	4.77E+76	−17.76	1.39E+05	760	
	3.59E+17	−0.56	1.05E+05	76,000	
R5 EF2 = EF2J + H	5.97E+79	−18.61	1.58E+05	0.76	[9] ^c
	7.13E+60	−13.04	1.48E+05	7.6	
	1.21E+41	−7.31	1.37E+05	76	
	9.02E+25	−2.96	1.27E+05	760	
R6 EF2 = EF24J + H	1.18E+80	−18.70	1.58E+05	0.76	[9] ^d
	1.50E+61	−13.14	1.48E+05	7.6	
	2.24E+41	−7.38	1.36E+05	76	
	1.29E+26	−3.00	1.27E+05	760	
R7 EF2 = EF25J + H	1.57E+80	−18.76	1.58E+05	0.76	[9] ^e
	1.15E+62	−13.41	1.48E+05	7.6	
	4.99E+42	−7.79	1.37E+05	76	
	3.59E+27	−3.43	1.28E+05	760	
H-Abstraction Reactions					
R8 EF2 + H = EF2J-A + H ₂	4.32E+00	3.98	1.38E+03		[50] ^f
R9 EF2 + H = EF2J-P + H ₂	2.88E+07	1.90	7.53E+03		[50] ^g
R10 EF2 + CH ₃ = EF2J-A + CH ₄	1.10E+12	0.00	9.10E+03		[51]
R11 EF2 + CH ₃ = EF2J-P + CH ₄	0.30E+00	4.00	8.20E+03		[51]
R12 EF2 + C ₃ H ₃ = EF2J-A + C ₃ H ₄ -a	1.60E+12	0.00	1.51E+04		[9]
R13 EF2 + C ₃ H ₃ = EF2J-A + C ₃ H ₄ -p	1.60E+12	0.00	1.51E+04		[9]
H-Transfer Reactions					
R14 EF2 = H45DE3O	2.58E+09	0.46	5.58E+04	7.6	Present work
	3.82E+09	0.64	5.95E+04	30	
	3.67E+09	0.79	6.18E+04	76	
	2.81E+09	1.05	6.58E+04	760	
	2.14E+09	1.15	6.70E+04	7600	
R15 EF2 = H23DE1O	6.39E+08	0.40	5.46E+04	7.6	Present work
	1.12E+09	0.57	5.79E+04	30	
	1.53E+09	0.69	6.03E+04	76	
	1.32E+09	1.02	6.56E+04	760	
	1.37E+09	1.15	6.79E+04	7600	
	H-Addition Reactions				
R16 EF2 + H = EF22H23J	1.93E+13	−1.68	−5.65E+03	0.76	[9] ^h
	3.41E+05	1.46	−7.24E+03	760	
	1.93E+06	1.33	−6.74E+03	1520	
R17 EF2 + H = EF23H22J	1.00E+09	1.44	4.33E+03		[9] ⁱ
R18 EF2 + H = EF24H25J	1.37E+05	2.50	3.79E+03		[9] ^j
R19 EF2 + H = EF25H24J	6.21E+28	−6.12	1.43E+03	0.76	[9] ^k
	4.37E+03	1.93	−9.28E+03	760	
	2.57E+02	2.36	−1.02E+04	1520	
R20 EF2 + H = Furan + C ₂ H ₅	1.20E+13	0.00	5.10E+03		[52] ^l
R21 EF2 + H = C ₅ H ₉ 1-1 + CO	1.30E+18	−1.24	8.36E+03	0.76	[9] ^m
	3.36E+19	−1.44	1.73E+04	760	
	4.31E+16	−0.64	1.64E+04	7600	
R22 EF2 + H = CH ₂ CHCHCO + C ₂ H ₅	1.34E+36	−6.42	2.06E+04		[53]
Secondary Reactions					
R23 MF2J = P34DE1O2J-C1	2.29E+01	3.01	3.26E+04	30	Present work
	2.67E+07	1.36	3.43E+04	760	
R24 EF2J-A = V2F + H	1.58E+12	−0.83	3.27E+04	7.6	Present work
	3.06E+12	−0.67	3.57E+04	30	
	3.83E+12	−0.54	3.79E+04	76	
	6.72E+12	−0.21	4.41E+04	760	
	3.53E+12	0.14	4.85E+04	7600	

(continued on next page)

Table 2 (continued)

Selected Reactions	A	n	E _a	Pressure (Torr)	Reference
R25 EF22J-P = V2F + H	2.31E+12	−0.97	2.20E+04	7.6	Present work
	1.59E+13	−1.09	2.27E+04	30	
	3.56E+14	−1.11	2.31E+04	76	
	3.58E+14	−1.16	2.51E+04	760	
	6.67E+14	−0.89	2.96E+04	7600	
R26 E2F2J-P = F2J + C ₂ H ₄	9.40E+12	−0.96	2.55E+04	30	Present work
	2.23E+14	−1.01	2.72E+04	760	
	2.09E+26	−4.85	2.25E+04	1520	
R27 EF22H23J = H3E1O2J-C1	9.01E+47	−12.51	2.94E+04	0.76	[9] ^a
	1.30E+26	−4.93	2.14E+04	760	
	2.09E+26	−4.85	2.25E+04	1520	
R28 EF25H24J = H5E3O4J-C1	1.52E+12	0.31	2.17E+04		[10] ^o
R29 V2F + H = Furan + C ₂ H ₃	2.21E+05	2.33	−1.18E+03		[53]
R30 V2F + CH ₃ = MF2 + C ₂ H ₃	1.20E+13	0.00	1.24E+04		[54]

Note:

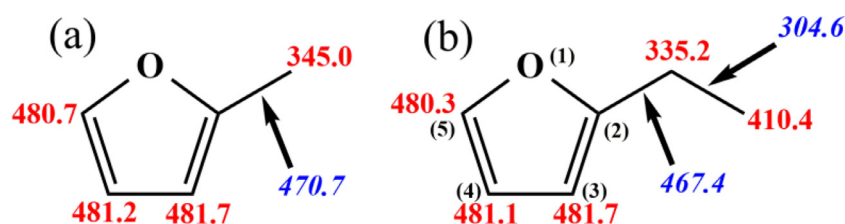
^a Referred to A1C₂H₅ = A1CHCH₃ + H.^b Referred to A1C₂H₅ = A1CH₂CH₂ + H.^c Referred to MF2 = MF23J + H.^d Referred to MF2 = MF24J + H.^e Referred to MF2 = MF25J + H.^f Referred to A1C₂H₅ + H = A1CHCH₃ + H₂.^g Referred to A1C₂H₅ + H = A1CH₂CH₂ + H₂.^h Referred to MF2 + H = MF22H.ⁱ Referred to MF2 + H = MF23H22J.^j Referred to MF2 + H = MF24H25J.^k Referred to MF2 + H = MF25H24J.^l Referred to C₈H₁₀ + H → C₆H₆ + C₂H₄.^m Referred to MF2 + H = C₄H₇1-1 + CO.ⁿ Referred to MF22H = P3E1O2J-C1.^o Referred to MF25H = P4E2O3J-C1.

Fig. 2. Molecular structures and bond dissociation energies (in kJ/mol) of (a) MF2 and (b) EF2. Normal text, C–H bond; italic, C–C bond. Numbers in parentheses are atom labels.

be compatible with the previous ones. The simulation of EF2 pyrolysis was carried out by the Plug Flow code in the Chemkin-PRO software [63]. The temperature profiles along the centerline of the flow reactor were used as the input parameters in the simulations. The reaction mechanism and thermodynamic data are provided in the *Supplementary Materials*.

4. Results and discussion

The structures and bond dissociation energies of MF2 and EF2 are shown in Fig. 2. A series of species were detected in EF2 pyrolysis experimentally, including radicals such as CH₃, C₃H₃ and cyclopentadienyl (C₅H₅), as well as stable species such as CO, methane (CH₄), ketene (CH₂CO), ethylene (C₂H₄), acetylene (C₂H₂), propene (C₃H₆), 2-vinylfuran (V2F), furan, CH₂CHCHCO and benzene (C₆H₆), etc. The major species mole fractions in the pyrolysis of EF2 and simulated results at both pressures are shown in Figs. 3–5. Some other species are shown in Fig. S2 in the *Supplementary Materials*. It can be observed that the formation temperatures and mole fractions of most species can be well reproduced by the current model. The current model was also verified by the pyrolysis data from Cheng et al. [6,7], as shown in Figs. S3–S4 in the *Supplementary Materials*. The performance of the present model also agreed well with that of Somers' model [9] and it may well reproduce the experimental mole fractions of some key species. The

rate of production (ROP) analysis and sensitivity analysis were conducted at 1212 K, 30 Torr and 1053 K, 760 Torr, where about 63% of the fuel was consumed and most of the species reach enough high mole fractions. The reaction networks of EF2 decomposition are drawn in Fig. 6(a). Corresponding sensitivity analysis of EF2 pyrolysis is shown in Fig. 7 to demonstrate the most sensitive reactions to EF2 consumption at 30 Torr and 760 Torr, respectively. As a key product, the sensitivity analysis of V2F is shown in Fig. 8 to explain the most sensitive reactions to its formation and consumption at both pressures. The reaction pathways that played important roles in the consumption of EF2 and comparison of EF2, MF2 and DMF25 pyrolysis are analyzed and discussed in the following sections.

4.1. Decomposition of EF2

Figure 3(a) shows the experimental and simulated results of EF2. Analogous to the pyrolysis of MF2 [7] and DMF25 [3], the ethyl side chain of EF2 is the active reaction center. The initial decomposition temperatures of EF2 are around 1025 K at 30 Torr and 875 K at 760 Torr. Compared with MF2 pyrolysis (around 1225 K at 30 Torr and 1025 K at 760 Torr [7]) and DMF25 pyrolysis (around 1180 K at 30 Torr and 1010 K at 760 Torr [6]) under the same pyrolysis conditions, the initial decomposition temperatures of EF2 are lower by about 200 K and 155 K at 30 Torr and 150 K and 135 K at 760 Torr than those of MF2 and DMF25, respectively.

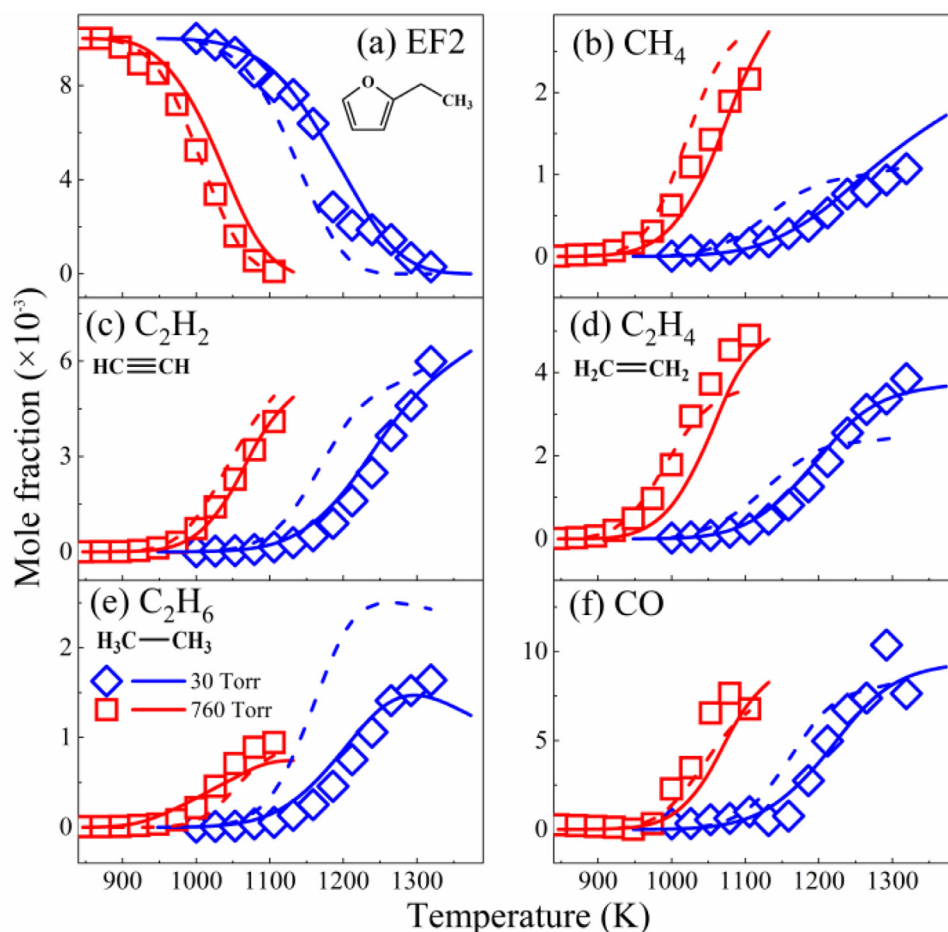


Fig. 3. Experimental and simulated mole fraction profiles of major species in 2-ethylfuran (EF2) pyrolysis. (a) 2-ethylfuran (EF2); (b) methane (CH_4); (c) acetylene (C_2H_2); (d) ethylene (C_2H_4); (e) ethane (C_2H_6) and (f) carbon monoxide (CO). Symbols, solid lines and dashed lines represent the experimental data, the simulated results by the current model, and the simulated results by Somers' model [9], respectively.

It is consistent with the theoretical results of the present work and that of Eldeeb and Akih-Kumgeh [24] and Xu et al. [25] that EF2 had much higher chemical reactivity than DMF25 and MF2. As shown in Fig. 2, the C–CH₃ bond is the weakest one with its low bond dissociation energy (BDE) of 304.6 kJ/mol in EF2. ROP analysis indicates that 44.4% of EF2 is consumed by R1, the breakage of C–CH₃ bond in the ethyl group, producing MF22J and CH₃, at 30 Torr. R1 was also the main formation pathway of MF22J and CH₃ at both pressures. As shown in Fig. 4(a), it could be seen that the mole fraction profile of CH₃ was slightly underestimated by Somers' model [9]. The rate constants of R1 were updated from the theoretical calculations in the current model. By this arrangement, the prediction for CH₃ is improved. The sensitivity analysis of EF2 pyrolysis in Fig. 7 indicates that R1 is the most sensitive reaction to the decomposition of EF2 at both pressures. Besides, pathways forming F2J and C₂H₅ through the C–CH₂ bond dissociation (R2) contributed negligibly to EF2 decomposition at both pressures due to the much higher BDE of C–CH₂ bond at 467.4 kJ/mol, comparing with that of C–CH₃ bond at 304.6 kJ/mol. Decomposition reactions of all C–H bonds had negligible contributions to the consumption of EF2 via R3–R7 due to their relatively higher BDEs, as seen in Fig. 2. Compared with the pyrolysis results of small furan series fuels, R1 is the most important unimolecular decomposition reaction in the pyrolysis of EF2. While in the pyrolysis of MF2, the breakage of the C–H bond in the methyl group only contributed around 2.0% at 30 Torr and 1.0% at 760 Torr to the consumption of MF2 [7] under the same fuel consumption rate. It can be concluded that the

weaker C–CH₃ bond leads to the decomposition of EF2 more readily, which may explain the fact that EF2 is more reactive in pyrolysis than furan and MF2.

Under pyrolysis conditions, H-abstraction reactions by H atom, CH₃ and C₃H₃ (R8–R13) consume a large proportion of EF2 according to ROP analysis. At 760 Torr, nearly 42.8% of EF2 is consumed via H-abstraction reactions. H-abstraction reactions may lead to the formation of E2F2J-A (R8, R10 and R12) and E2F2J-P (R9, R11 and R13), as well as the accompanying products of hydrogen (H₂), CH₄, allene (C₃H₄-a) and propyne (C₃H₄-p), respectively. The rate constants of R8–R11 were updated from the ethylbenzene model [50] and toluene model [51] to replace the estimated values in Somers' model [9]. By this arrangement, the predictions for EF2 are in better agreement with the experimental results, especially at 30 Torr, as shown in Fig. 3(a). R8 and R9 contributed nearly 19.7% and 22.5% to EF2 consumption at 30 Torr and 760 Torr according to ROP analysis, respectively. R10 and R11 contributed 4.3% and 11.4% at 30 Torr and 760 Torr, respectively. H-abstraction reactions by H (R8 and R9) account for more portion of EF2 consumption than those by CH₃ and C₃H₃ (R10–R13). E2F2J-A is the prior H-abstraction product of EF2 due to the lower BDE of allylic C–H bond.

As for H-transfer reactions (R14–R15), their contributions to the consumption of EF2 are negligible according to the ROP analysis. This conclusion is the same as the simulation results by Somers' model [9]. Direct carbene-mediated decomposition species were also not detected in the pyrolysis of EF2.

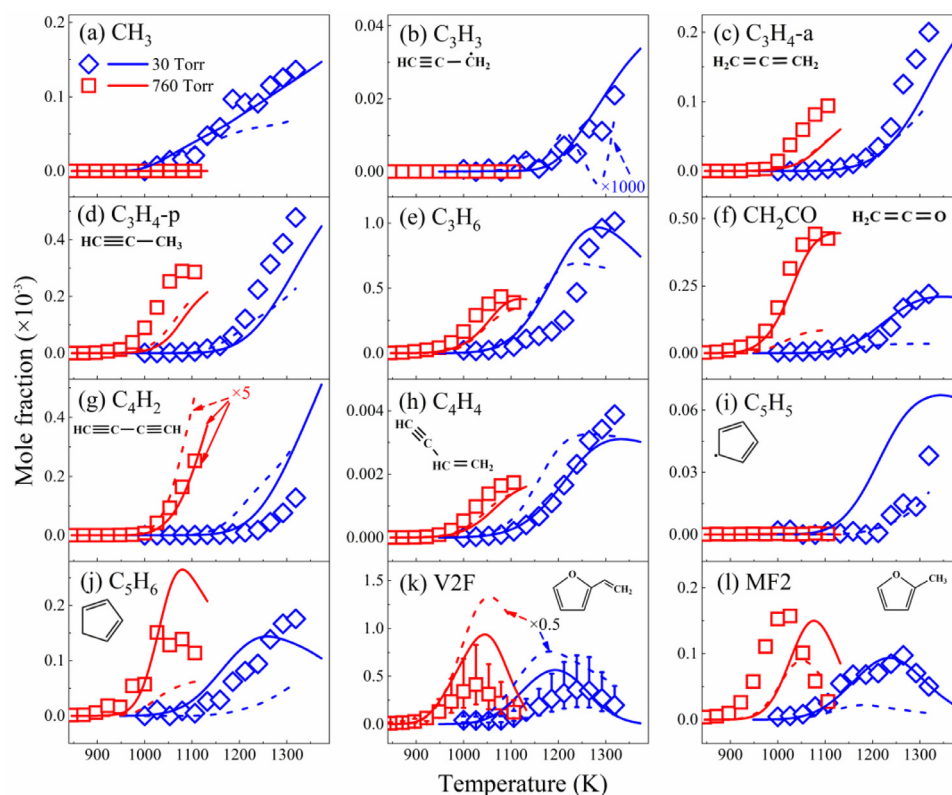


Fig. 4. Experimental and simulated mole fraction profiles of important species in EF2 pyrolysis. (a) methyl (CH_3), (b) propargyl (C_3H_3), (c) allene ($\text{C}_3\text{H}_4\text{-a}$), (d) propyne ($\text{C}_3\text{H}_4\text{-p}$), (e) propylene (C_3H_6), (f) ketene (CH_2CO), (g) diacetylene (C_4H_2), (h) vinylacetylene (C_4H_4), (i) cyclopentadienyl (C_5H_5), (j) 1,3-cyclopentadiene (C_5H_6), (k) 2-vinylfuran (V2F) and (l) 2-methylfuran (MF2). Symbols, solid lines, and dashed lines represent the experimental data, the simulated results by the current model, and the simulated results by Somers' model [9], respectively.

H-addition reactions are another key group of reactions for the consumption of EF2 (R16–R22). Among the four possible H-addition positions, C(2) and C(5) are more favorable positions than C(3) and C(4). H-addition on C(2) leads to the production of EF22H23J (R16), which decomposes to furan + C_2H_5 (R20) or $\text{C}_5\text{H}_9\text{1-1} + \text{CO}$ (R21), with contributions around 5.4% or 8.1% at 30 Torr and 8.7% or 2.9% at 760 Torr to EF2 consumption, respectively. Similar to EF22H23J, H-addition on C(5) leads to the production of EF25H24J (R19), which decomposes to $\text{CH}_2\text{CHCHCO} + \text{C}_2\text{H}_5$ (R22), with a contribution around 15.7% and 23.1% to EF2 consumption at low and atmospheric pressures, respectively.

As shown in Fig. 7, it can be seen that R1 is the most key reaction in the pyrolysis of EF2 at 30 Torr and 760 Torr. H-abstraction reaction on the methylene group (R8) and H-addition reaction on C(5) (R19) follow R1 in contributing to the consumption of EF2, although with much lower importance.

4.2. Further decomposition of primary decomposition products

As shown in the previous section, the products of unimolecular decomposition reactions of EF2 are mainly MF22J and CH_3 . Based on ROP analysis, MF22J is a significant intermediate product of EF2 pyrolysis, especially at 30 Torr. The main consumption pathway of MF22J is the $\beta\text{-C-C}$ scission to generate 3,4-pentadienal-2-yl (P34DE102J-C1) (R23), which eventually decomposes to 1,3-butadien-1-yl ($\text{C}_4\text{H}_5\text{-n}$) + CO eventually, as shown in Fig. 6(a). This reaction pathway is the dominant generation pathway of $\text{C}_4\text{H}_5\text{-n}$ and CO. As for $\text{C}_4\text{H}_5\text{-n}$, it could be decomposed to produce vinylacetylene (C_4H_4) + H (R31) or C_2H_2 + vinyl (C_2H_3) and R31 is the dominated generation pathway of C_4H_4 at both pressures. As shown in Fig. 4(h), it could be seen that the mole fraction profile of C_4H_4 was slightly overpredicted by Somers' model [9]. The rate

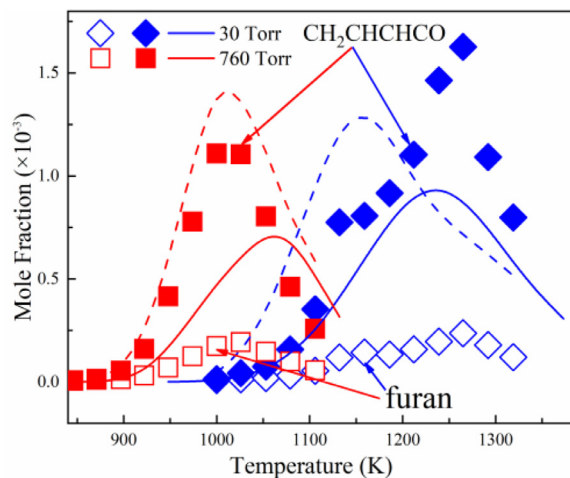


Fig. 5. Experimental and simulated mole fraction profiles of the isomer pair furan/vinylketene (CH_2CHCHCO). Symbols, solid lines and dashed lines represent the experimental data, the simulated results by the current model, and the simulated results by Somers' model [9], respectively. Note that the experimental data represent the upper and lower limits of the corresponding species, respectively.

constants of R23 were recomputed in this work. By the arrangement, the predictions for C_4H_4 are slightly improved, especially at low pressure. Besides, R31 also plays a key role in the consumption of EF2 due to the generation of H atom at 30 Torr, as shown in Fig. 7. MF22J may also be converted to MF2 by combining an H atom, which dominates the generation of MF2 at 30 Torr according to ROP analysis. However, due to the high reactivity of MF22J and obvious quenching effect, the accurate qualification of

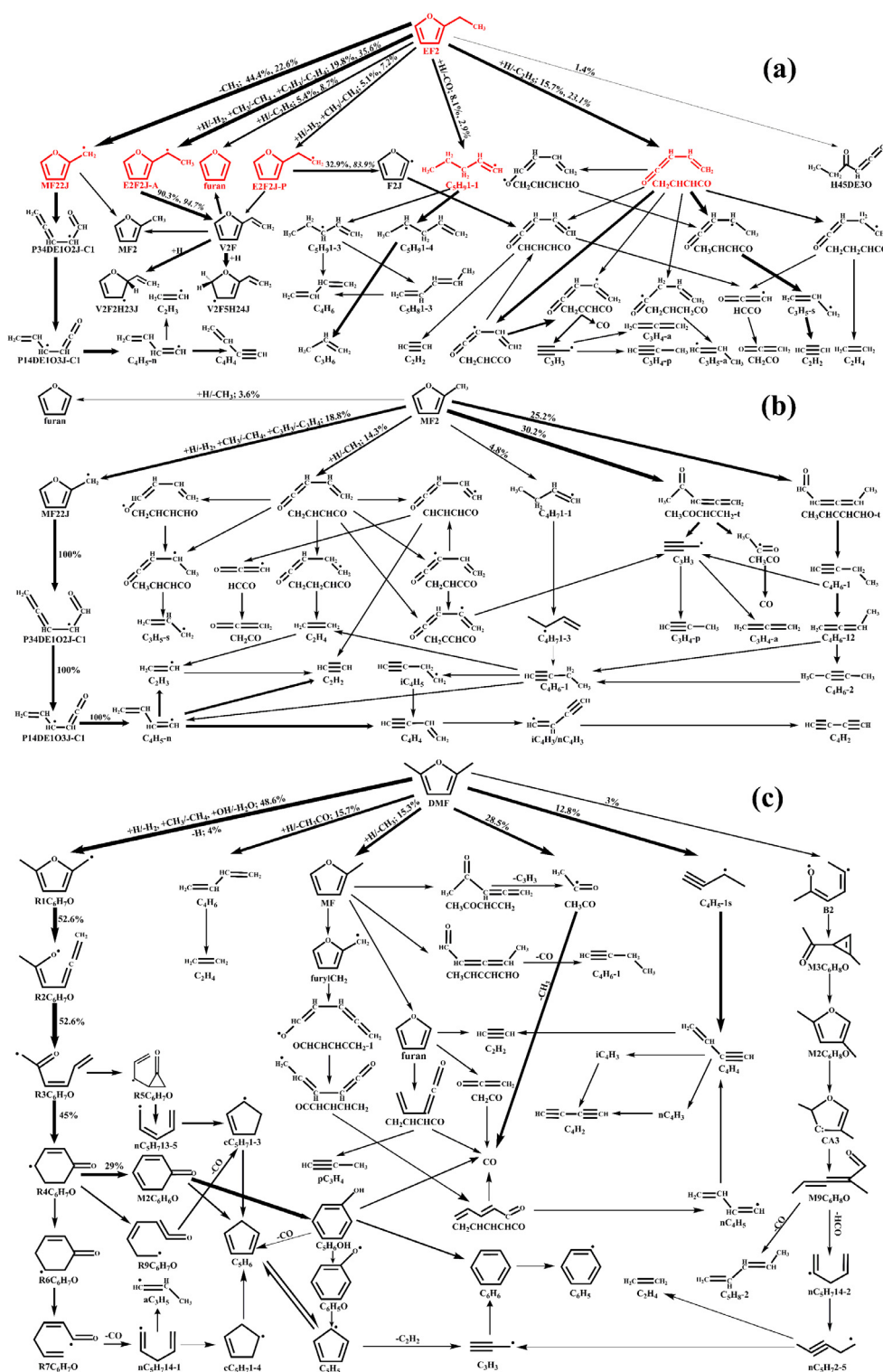
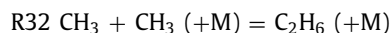
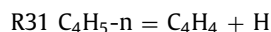


Fig. 6. Pyrolysis reaction networks of (a) EF2 at 1212 K, 30 Torr (normal text) and at 1053 K, 760 Torr (italic text), (b) MF2 at 1390 K and 30 Torr and (c) DMF25 at 1346 K and 30 Torr. The numbers reflect the contribution of the corresponding reaction pathways in total reaction flux. All of the flux analyses correspond to about 63% of fuel consumption.

MF22J can't be realized in this work. As for the accompanying product of MF22J, CH_3 mainly goes self-combination reaction (R32) to produce ethane (C_2H_6) at both low and atmospheric pressures. Somers' model slightly overpredicted C_2H_6 , as shown in Fig. 3(e). In current work, the rate constants of R32 from GRI [64] were adopted. It can be observed that the prediction for C_2H_6 is in better agreement with the experimental results in this work. R32

is also the primary formation pathway of C_2H_6 in EF2 pyrolysis at 30 Torr.



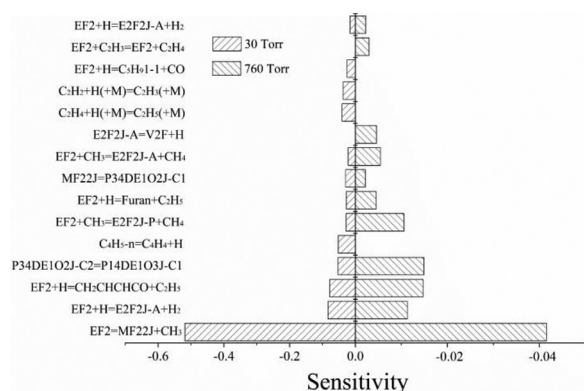


Fig. 7. Sensitivity analysis of EF2 consumption at 1212 K, 30 Torr and at 1053 K, 760 Torr, corresponding to 63% of fuel consumption.

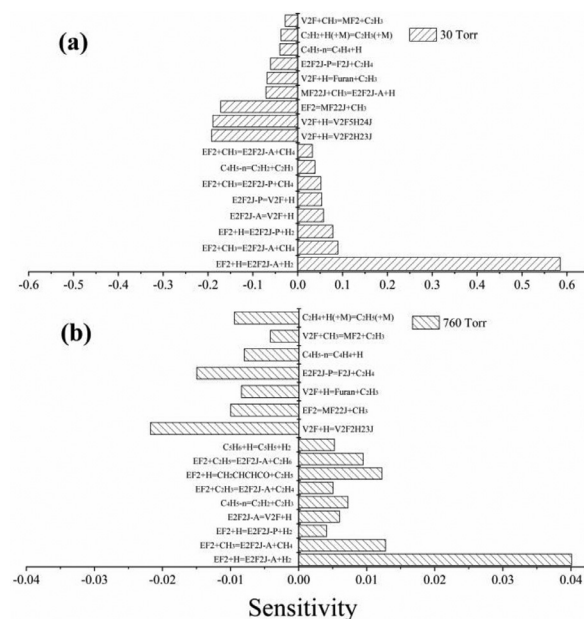
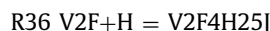
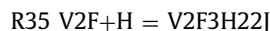
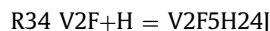
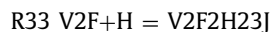


Fig. 8. Sensitivity analysis of V2F formation and consumption at (a) 1212 K, 30 Torr and (b) 1053 K, 760 Torr, corresponding to 63% of fuel consumption.

H-abstraction reaction of EF2 will lead to the formation of E2F2J-A and E2F2J-P. E2F2J-A and E2F2J-P both dominantly decompose to V2F via β -C-H scissions (R24–R25) at 30 Torr according to ROP analysis. These two reactions contribute nearly 100% to the production of V2F at both pressures. At 760 Torr, E2F2J-P is mainly consumed via β -C-C cleavage (R26) to generate F2J and C_2H_4 , accounting for about 83.9% of its consumption. As is shown in Fig. 8, the sensitivity analysis of V2F was conducted at 1212 K, 30 Torr and 1053 K, 760 Torr, respectively. It could be observed that the H-abstraction reaction of EF2 by H atom (R8) has the maximum sensitivity coefficient to V2F formation at 30 Torr and 760 Torr. This point is in accordance with the simulation results that the main product of the same reaction, E2F2J-A, is the source of the most sensitive reaction directly leading to the production of V2F (R24). According to ROP analysis, about 77.5% and 71.0% of V2F is consumed via H-addition reactions on C(2), R33, or C(5), R34, at 30 Torr and 760 Torr, respectively. Besides, 11.5% and 17.8% of V2F is converted to furan and C_2H_3 (R29) via *ipso*-substitution of H atom at both low and atmospheric pressures, respectively. By adding H-addition reactions and *ipso*-substitution reaction by H atom of V2F into Somers' model, the prediction of V2F is improved substantially in the current work, as seen in Fig. 4(k). By

the same rationale, *ipso*-substitution of V2F by CH_3 (R30) to produce MF2 is also possible. It can be seen that the prediction of MF2 is improved obviously, as shown in Fig. 4(l). In the present work, the rate constants of R30 were taken from the similar reactions of styrene ($\text{C}_6\text{H}_5\text{C}_2\text{H}_3$) model [54]. The rate constants of R33–R36 were taken from Somers' model [9]. According to the ROP analysis, about 19.0% and 28.9% of furan are generated by R29 at 30 Torr and 760 Torr, respectively, while 22.7% and 44.5% of MF2 are generated by R30 at 30 Torr and 760 Torr, respectively.



As for the products of H-addition reactions of EF2, the reaction leading to the formation of CH_2CHCHCO , R22, contributes over 15.0% in the pyrolysis of EF2. This case is similar to that in MF2 pyrolysis. As a key product of H-addition reactions of MF2, it was the theoretical work of Somers et al. that first posited this pathway and was then detected in MF2 flames [65]. CH_2CHCHCO is also detected in the present work. According to ROP analysis, about 71.3% and 27.1% of CH_2CHCHCO is consumed via the H-addition reactions to produce ethylketen-3-yl ($\text{CH}_3\text{C}^*\text{HCHCO}$), ethylketen-4-yl ($\text{C}^*\text{H}_2\text{CH}_2\text{CHCO}$), allylcarbonyl ($\text{CH}_2\text{CHCH}_2\text{C}^*\text{O}$) and 3-butenal-1-yl ($\text{CH}_2\text{CHCHCHCO}^*$) at 30 Torr and 760 Torr, respectively. The dominant consumption reactions of these four species are either β -C-C scission reactions or direct releasing a CO molecule. Decomposition of $\text{CH}_3\text{C}^*\text{HCHCO}$ leads to the formation of 1-propenyl ($\text{CH}_2\text{CHC}^*\text{H}_2$, $\text{C}_3\text{H}_5\text{-s}$), which mainly decomposes to produce C_2H_2 and CH_3 . This reaction pathway dominates the formation of C_2H_2 at both 30 Torr and 760 Torr. Subsequent decomposition of $\text{C}^*\text{H}_2\text{CH}_2\text{CHCO}$ leads to the generation of $\text{C}_2\text{H}_4 + \text{HCCO}$, while the decomposition of $\text{CH}_2\text{CHCH}_2\text{C}^*\text{O}$ leads to allyl ($\text{CH}_3\text{CHC}^*\text{H}$, $\text{C}_3\text{H}_5\text{-a}$) + CO. As to the reaction pathway of $\text{CH}_2\text{CHCHCHCO}^*$, it mainly produces CH_3CHCHCO by isomerization reactions. Subsequent dominant decomposition pathway of CH_3CHCHCO is to generate 1-propenyl ($\text{C}_3\text{H}_5\text{-s}$) + CO. About 25.9% and 6.5% of CH_2CHCHCO is consumed via H-abstraction reactions, mainly leading to the formation of vinylketen-2-yl ($\text{CH}_2=\text{CH}-\text{C}^*=\text{CO}$, $\text{CH}_2\text{CHC}^*\text{CO}$) at both 30 Torr and 760 Torr, respectively. By isomerization, $\text{CH}_2\text{CHC}^*\text{CO}$ dominantly transforms to vinylketen-3-yl ($\text{CH}_2=\text{C}^*-\text{CH}=\text{CO}$, $\text{CH}_2\text{C}^*\text{CHCO}$) at 30 Torr and 760 Torr. Further decomposition reactions of $\text{CH}_2\text{C}^*\text{CHCO}$ could produce $\text{C}_3\text{H}_3 + \text{CO}$. This reaction pathway is the most key production pathway of C_3H_3 at 30 Torr and 760 Torr. As is shown in Fig. 4(b), it could be seen that the prediction of C_3H_3 is much improved by the current model.

Besides, it also should be pointed out that H-addition reactions leading to CH_2CHCHCO (R22) competed with *ipso*-substitution reaction leading to furan, R20. Unfortunately, the main products of these two competing reactions, CH_2CHCHCO and furan, are a pair of isomers at $m/z = 68$. To worsen the case, mole fractions of the two species cannot be distinguished by the method of SVUV-PIMS due to their close ionization energies. Thus, all species at $m/z = 68$ are treated either as furan or CH_2CHCHCO in convoluting the experimental mole fractions. Correspondingly, the experimental results should be seen as the lower and upper limits for the mole fraction of the respective isomers, i.e., neither experimental mole fractions of the two species should be higher than the maximum values and lower than the minimum values shown in Fig. 5. Still, it should be pointed out that PICS of CH_2CHCHCO was obtained by analogy with allylketene ($\text{CH}_2=\text{C}=\text{CH}-\text{CH}=\text{C}=\text{O}$), as described in recent works [20,66].

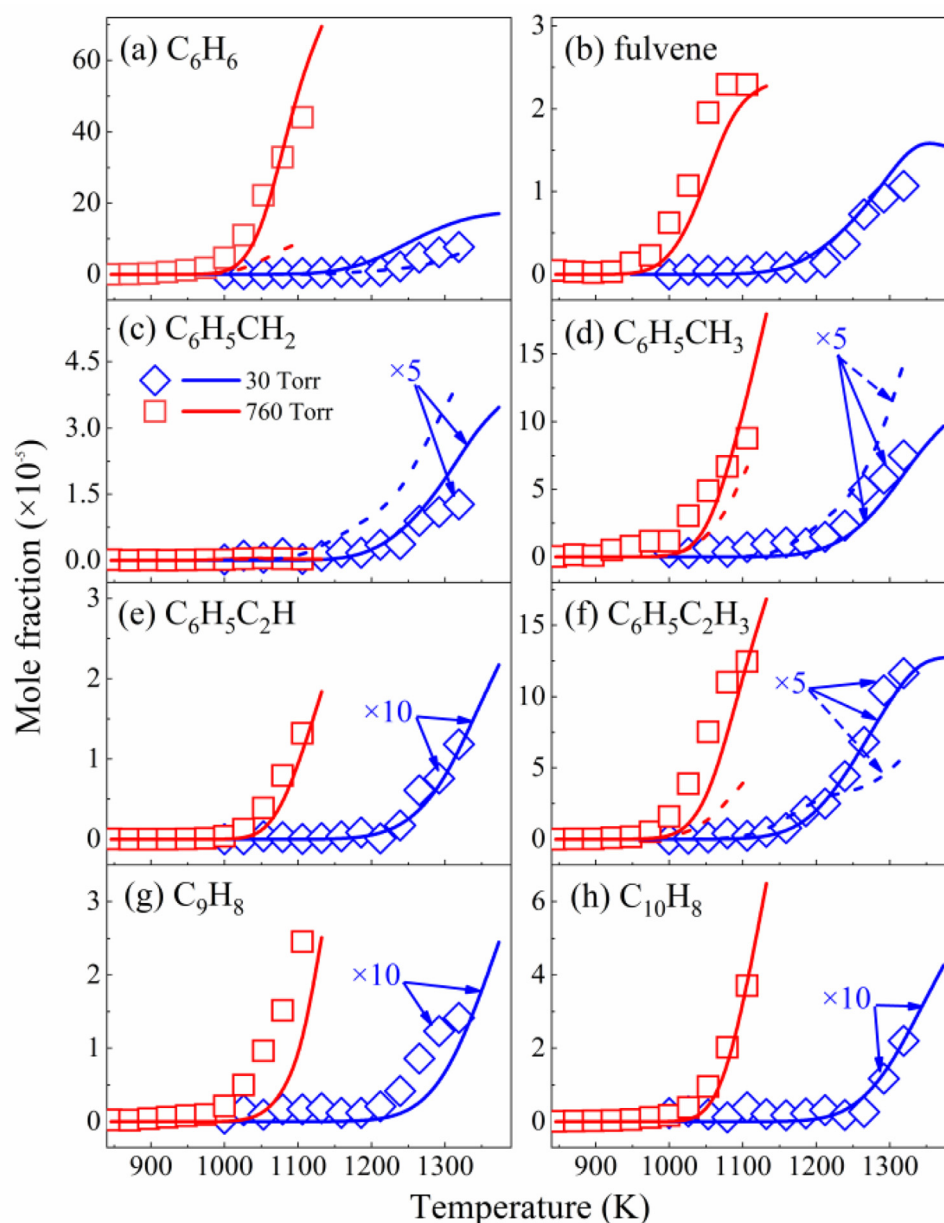


Fig. 9. Experimental and simulated mole fraction profiles of (a) benzene (C_6H_6), (b) fulvene, (c) benzyl radical ($C_6H_5CH_2$), (d) toluene ($C_6H_5CH_3$), (e) phenylacetylene ($C_6H_5C_2H$), (f) styrene ($C_6H_5C_2H_3$), (g) indene (C_9H_8) and (h) naphthalene ($C_{10}H_8$). Symbols, solid lines, and dashed lines represent the SVUV-PIMS data, the simulated results by the updated model, and the simulated results by the Somers' model [9], respectively.

4.3. Formation of aromatics in EF2 pyrolysis

Figure 9 shows the experimental and simulated results of eight aromatic hydrocarbons in the present work, including C_6H_6 , fulvene, benzyl radical ($C_6H_5CH_2$), toluene ($C_6H_5CH_3$), phenylacetylene ($C_6H_5C_2H$), $C_6H_5C_2H_3$, ethylbenzene ($C_6H_5C_2H_5$), indene (C_9H_8), naphthalene ($C_{10}H_8$). It can be seen that the current model well reproduced the formation temperatures and mole fractions of these aromatics. The dominant production pathways are demonstrated through ROP analyses and will be discussed in the following section.

C_6H_6 is the most key aromatic species in EF2 pyrolysis, with its maximum yields and its key effect on the formation of other aromatic products. The production of C_6H_6 is dominated by the reaction of $C_4H_4 + C_2H_2$ at both low and atmospheric pressures according to ROP analysis in the current model, with the rate constants of this reaction updated from the allene and propyne model

of Hansen et al. [67]. This arrangement also improved the prediction for C_6H_6 . In Somers' model, the production of C_6H_6 is dominated by the combination reaction of C_4H_5 -n and C_2H_2 according to ROP analysis. Fulvene, an isomer of C_6H_6 , was added into the present model, referring to the MF2 model of Cheng et al. [7]. ROP analysis shows that the self-combination reaction of C_3H_3 dominated the generation of fulvene at both pressures, with rate constants of this reaction taken from the allene and propyne model [67]. As shown in Fig. 9(b), the simulated mole fraction profiles of fulvene by the current model agreed well with the experimental results.

For the ordinary C7 aromatics series, $C_6H_5CH_2$ and $C_6H_5CH_3$ were detected and quantified in the present work. As shown in Fig. 9(c) and (d), the simulated and experimental mole fraction profiles of $C_6H_5CH_2$ and $C_6H_5CH_3$ well reproduce the experimental results, respectively. It can be observed that the current model slightly improves the prediction of the two aromatics

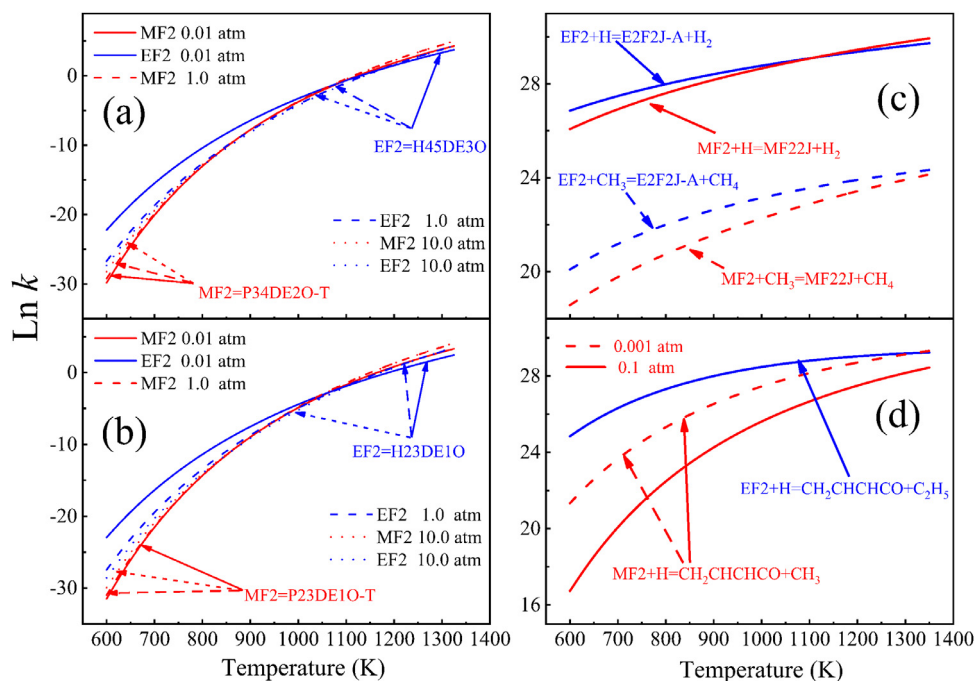


Fig. 10. Comparison of rate constants of major H-transfer reactions, (a) and (b), H-abstraction reactions, (c), and H-addition reactions, (d), in EF2 and MF2 pyrolysis.

and agrees well with the experimental mole profiles. The production of $C_6H_5CH_2$ is dominated by the combination reaction of $C_5H_5 + C_2H_2$ at 30 Torr and 760 Torr in the present model according to ROP analysis. In Somers' model, the combination reaction of $C_3H_3 + C_4H_4$ dominated the formation of $C_6H_5CH_2$, especially at 30 Torr. For $C_6H_5CH_3$, it could be mainly generated by the combination reactions of $C_6H_5 + CH_3$, $C_6H_5CH_2 + H$ and $C_3H_3 + 1,3$ -butadiene (C_4H_6) at both pressures.

For C8 aromatics, $C_6H_5C_2H$ and $C_6H_5C_2H_3$ were detected and quantified in the present work, as shown in Fig. 9(e) and (f). It could be seen that the current model well predicted the experimental results of $C_6H_5C_2H$ and $C_6H_5C_2H_3$. The production of $C_6H_5C_2H$ is dominated by the combination of phenyl (C_6H_5) + C_2H_2 according to ROP analysis. As for $C_6H_5C_2H_3$, the combination reaction of $C_5H_5 + C_3H_3$ dominates its generation at both pressures according to ROP analysis. In Somers' model, the production of $C_6H_5C_2H_3$ was dominated by the combination reaction of $C_4H_5 + C_4H_4$. As can be seen from Fig. 9(f) that the current model improved prediction of the mole fraction profile of $C_6H_5C_2H_3$ with the adoption of the related sub-mechanism from Cheng et al. [7].

As for C9 and C10 aromatics, C_9H_8 and $C_{10}H_8$ were detected and quantified in the present work. As shown in Fig. 9(g) and (h), it can be observed that the present model well reproduced the formation temperatures and mole fractions of these two aromatics. The production of C_9H_8 is dominated by combination reactions of $C_6H_5 + C_3H_3$ and $C_6H_5CH_2 + C_2H_2$, while the production of $C_{10}H_8$ is dominated by the combination reaction of $C_5H_5 + C_5H_5$ and $C_6H_5 + C_4H_4$ at 30 Torr and 760 Torr according to ROP analysis.

4.4. Comparison of MF2, EF2 and DMF25 pyrolysis

Pyrolysis flux analysis of MF2 at 1390 K, 30 Torr and that of DMF25 at 1346 K, 30 Torr, corresponding to 63% fuel consumption, are also displayed in Fig. 6 for comparison, with MF2 pyrolysis mechanism from Cheng et al. [7] and DMF25 pyrolysis mechanism from Cheng et al. [6]. At 30 Torr, it could be seen that the H-transfer reaction pathways contributed 1.4%, 55.4% and 12.8% fuel consumption in EF2, MF2 and DMF25 pyrolysis, respectively. Comparisons of the rate constants of the H-transfer reactions for MF2

and EF2 are shown in Fig. 10(a) and (b). The rate constants of H-transfer from C(4) to C(5), i.e., R14, in EF2 pyrolysis are higher than those of the corresponding reactions in MF2 pyrolysis below 1050 K. While above 1050 K, these two rate constants are very close under the three pressures. As for H-transfer from C(3) to C(2), i.e., R15, the trend is similar for these two fuels. The computed parameters of these H-transfer reactions for EF2 in this work are at the same level of those proposed by analogy in Somers' model [9] for EF2. This huge gap between the contribution of the two similar reaction pathways may be attributed to the much faster C-CH₃ cleavage in the decomposition of EF2, dwarfing the contribution of H-transfer reactions. While in DMF25, the modest contribution of H-transfer is dwarfed by the H-abstraction reactions, with a contribution of 48.6%.

Unimolecular decomposition reactions accounted for negligible fuel consumption in MF2 pyrolysis and 3.8% in DMF25 pyrolysis at 30 Torr, by breakage of C-H bond on the CH₃ side chain(s). While that value increased dramatically to 44.4% in EF2 pyrolysis, at this time by breakage of C-C bond on the C₂H₅ side chain. This difference is reasonable since in EF2 pyrolysis, breakage of C-H bond on the C₂H₅ side chain also contributes negligibly to the consumption of EF2. The enhanced contribution of unimolecular decomposition in EF2 pyrolysis is attributed to the weaker C-CH₃ bond (304.6 kJ·mol⁻¹) in EF2. The higher BDEs of C-H bonds on CH₃ side chains resulted in the fewer consumption of DMF25 (346 kJ·mol⁻¹) and MF2 (345 kJ·mol⁻¹) by directly breaking C-H bonds. However, the double branch chains of CH₃ structure in DMF25 may account for the slightly higher contribution by the direct unimolecular decomposition reactions than in MF2.

H-abstraction reactions accounted for 24.9% (19.8% at CH₂ + 5.1% at CH₃), 18.8% and 48.6% fuel consumption in EF2, MF2 and DMF25 pyrolysis, respectively. The fact that H-abstraction reactions contribute higher in EF2 pyrolysis than in MF2 reflects the difference between the two side chains in the two fuels, since the larger C₂H₅ side chain in EF2 than CH₃ in MF2 incurred more attacks from the radical pool. It is consistent with the rate constants of H-abstraction reactions at the CH₂ position, as shown in Fig. 10(c), with the parameters proposed in ethylbenzene mechanism [50]. DMF25 shows much higher percentage by H-abstraction

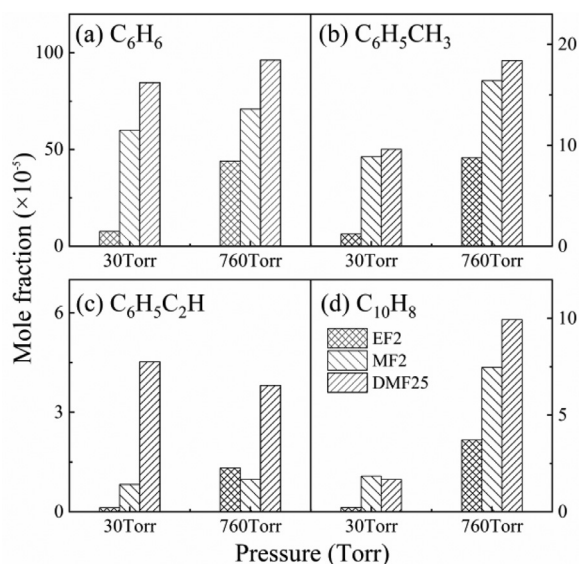


Fig. 11. Comparison of mole fractions of (a) benzene (C_6H_6), (b) toluene ($C_6H_5CH_3$), (c) phenylacetylene ($C_6H_5C_2H$) and (d) naphthalene ($C_{10}H_8$) at 98% fuel conversion in EF2 pyrolysis (experimental results), MF2 pyrolysis (simulated results by Cheng's mechanism [7]) and DMF25 pyrolysis (simulated results by Cheng's mechanism [6]) with the same fuel inlet mole fraction (1%) at 30 and 760 Torr.

reaction with its two CH_3 side chains and hence enhanced overall H-abstraction reaction possibilities, as demonstrated in references [6,7].

H-addition reactions, including *ipso*-substitutions, to C(2) or C(5) had similar reaction pathways for EF2, MF2 and DMF25. It can be seen that the H-addition reactions accounted for 29.2% (15.7% + 8.1% + 5.4%), 22.7% (14.3% + 4.8% + 3.6%) and 31% (15.7% + 15.3%) fuel consumption in EF2, MF2 and DMF25 pyrolysis, respectively, according to ROP analysis in Fig. 6. Figure 10(d) shows the comparison of the rate constants of H-addition on C(2), i.e., R22 ($EF_2 + H = CH_2CHCHCO + C_2H_5$) and R37 ($MF_2 + H = CH_2CHCHCO + CH_3$). It can be seen that the rate constant of R22 is much higher than that of R31. Thus, contribution of R22 to EF2 consumption should be higher than that of its counterpart, R37 to MF2 consumption. Again, due to the dwarfing effect of C–C bond cleavage on the C_2H_5 side chain, the final contribution of R22 to EF2 consumption is comparable with that of R37 to MF2 consumption.



Figure 11 shows the experimental mole fractions of C_6H_6 , $C_6H_5CH_3$, $C_6H_5C_2H$ and $C_{10}H_8$ in EF2 pyrolysis in this work with the simulated ones in MF2 and DMF25 pyrolysis using MF2 [7] and DMF25 [6] mechanisms of Cheng et al., at the same conversion rate of 98%, where the fuel was almost completely consumed and most of the aromatic products reached higher mole fractions. Obviously, the mole fractions of C_6H_6 and $C_6H_5CH_3$ are unanimously lower in EF2 pyrolysis than those in MF2 and DMF25 pyrolysis at both pressures. This trend indicates that EF2 should have a lower tendency to generate soot precursors than MF2 and DMF25 [68]. Figure 12 shows the production networks of C_6H_6 and $C_{10}H_8$ at 98% fuel conversion in the pyrolysis of EF2, MF2 and DMF25 with the same fuel inlet mole fraction (1%) at 30 Torr. It can be seen that C_6H_6 and $C_{10}H_8$ can be produced from the reactions of $C_3H_3 + C_3H_3$ (R38), $C_4H_4 + C_2H_3$ (R39) and $C_5H_5 + C_5H_5$ (R40) in EF2, MF2 and DMF25 pyrolysis. In the pyrolysis of EF2, the generation of C_6H_6 was mainly dominated by R39. The flux analysis of EF2 in Fig. 12(a) shows that C_4H_4 was mainly produced by a series of

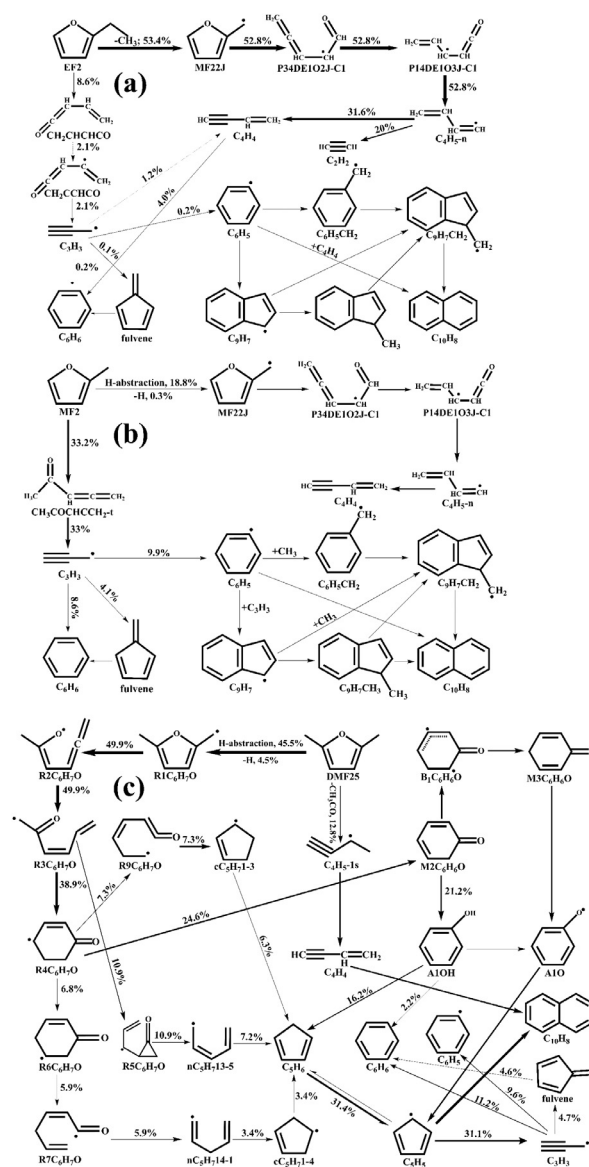


Fig. 12. Production networks of C_6H_6 and $C_{10}H_8$ at 98% fuel conversion in (a) EF2 pyrolysis, (b) MF2 pyrolysis and (c) DMF25 pyrolysis with the same fuel inlet mole fraction (1%) at 30 Torr.

direct decomposition reactions of EF2. This reaction pathway accounts for 4.0% fuel consumption to produce C_6H_6 in the pyrolysis of EF2. C_3H_3 was generated by the decomposition of CH_2CCHCO , a reaction pathway accounting for only 0.3% of fuel consumption to generate C_6H_6 in the pyrolysis of EF2. Thus, a total of 4.3% (4.0% + 0.3%) fuel is pyrolyzed to produce C_6H_6 . While the production of $C_{10}H_8$ was dominated by the combination reaction of C_6H_5 and C_4H_4 , the combination reaction of C_3H_3 dominates the production of C_6H_5 . About 0.2% of EF2 was consumed to produce C_6H_5 . As a result, only a small amount of $C_{10}H_8$ was detected and quantified in EF2 pyrolysis. Based on ROP analysis of EF2 pyrolysis as shown in Fig. 6(a), the unique decomposition pathway of EF2 to generate V2F occupied a large proportion of EF2 with its C_2H_5 side chain. Consumption of V2F mainly leads to the formation of some stable species, i.e., C_2H_2 . This reaction pathway competes the consumption of EF2 and thus leads to the lower production of C_6H_6 and $C_{10}H_8$ in EF2 pyrolysis. While in the similar cases of MF2 pyrolysis, R38, i.e., combination of C_3H_3 , plays a vital effect in the formation of C_6H_6 and $C_{10}H_8$. It should be pointed out that C_3H_3 was largely generated from the direct decomposition of $CH_3COCHCH_2-$

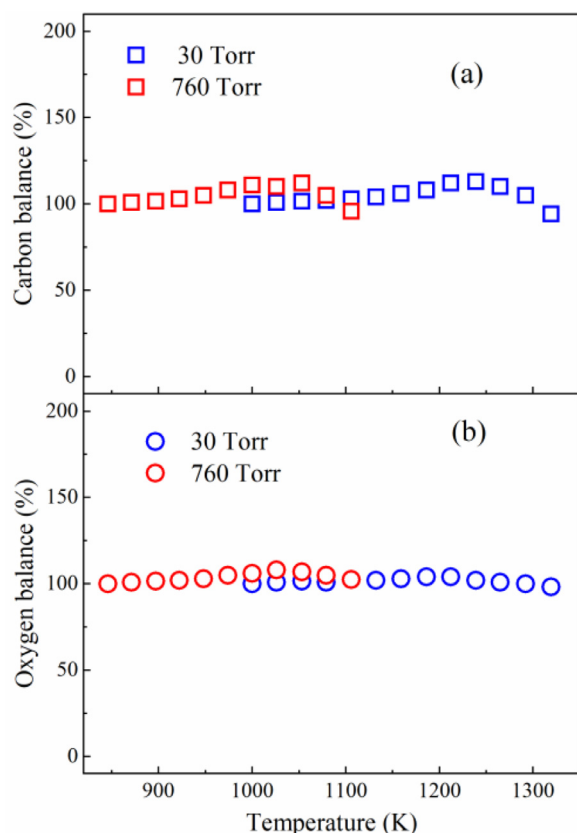


Fig. 13. Carbon balance (a) and oxygen balance (b) distribution in the pyrolysis of EF2 at 30 Torr and 760 Torr, respectively.

t according to the flux analysis, as shown in Fig. 12(b). Contribution of R39, i.e., reaction of $C_4H_4 + C_2H_3$, contributes negligibly to the formation of C_6H_6 . As a result, 12.7% (8.6% + 4.1%) and 9.9% of MF2 was consumed to generate C_6H_6 and C_6H_5 at 30 Torr in the pyrolysis, respectively. Besides, the overall mole fractions of C_3H_3 , C_4H_4 and C_6H_6 in EF2 pyrolysis were one order of magnitude higher than those in EF2 pyrolysis. As for the case in DMF25 pyrolysis [6], C_5H_5 radical played the most important role in the formation of C_6H_6 and $C_{10}H_8$, as shown in Figs. 6(c) and 12(c). The unique structure of DMF25 easily leads to the production of a large number of C_5H_5 radicals and hence the enhanced contribution of R38 and R40 to the production of C_6H_6 and $C_{10}H_8$.

If EF2 is proved to produce fewer aromatic species and even fewer soot precursors in its pyrolysis and combustion, effort in mass-producing biomass-based furan series fuels should focus on such a candidate in the future.

4.5. Carbon and oxygen balances in pyrolysis of EF2

According to the set-up of the experiments, the mole fractions of carbon atoms (number of carbon atoms divided by the total numbers of carbon, hydrogen, oxygen, Ar and Kr atoms) and oxygen atoms should be 0.06 and 0.01, respectively. The mole fractions of carbon atoms and oxygen atoms as summarized by the experimental results were shown in Fig. 13 at 30 and 760 Torr. The initial carbon balance and oxygen balance were quantified as 100%. By dividing the experimental mole fractions of total carbon atoms with 0.06, the carbon balance was displayed as a function of experimental temperature. Basically, the carbon balance was maintained at a reasonable level, within the experimental uncertainties. Oxygen balance follows a similar trend. No major species were omitted in detecting and measuring.

5. Conclusion

Flow reactor pyrolysis of 2-ethylfuran (EF2) was studied at pressures of 30 and 760 Torr and at temperatures of 846 – 1319 K by using synchrotron vacuum ultraviolet photoionization mass spectrometry. Several radicals and stable species in EF2 pyrolysis were detected and quantified, including methyl, propargyl, vinylketene, furan, 2-methylfuran, 2-vinylfuran, benzene, toluene, and naphthalene, etc. A comprehensive EF2 pyrolysis model was updated and validated against the experimental data. The present model could reproduce the results of most pyrolysis species within the experimental uncertainty at both low and atmospheric pressures. The rate of production analysis and sensitivity analysis were conducted to analyze the main reaction pathways in EF2 pyrolysis. It could be concluded that the unimolecular decomposition to produce 2-furylmethyl, H-abstraction by H-atom and methyl to produce 1-(2-furyl)ethyl, and H-addition reactions to produce vinylketene, furan and 1-penten-1-yl contributed overwhelmingly to EF2 consumption at both pressures in EF2 pyrolysis. In this work, the ethyl side chain of EF2 led to a very key C–CH₃ bond dissociation pathway in the decomposition of the fuel, a unique decomposition pathway compared with 2-methylfuran and 2,5-dimethylfuran. 2-Vinylfuran, 2-furylmethyl and vinylketene were key species in EF2 pyrolysis. Aromatic products were observed with lower concentration levels in EF2 pyrolysis compared with those in 2-methylfuran and 2,5-dimethylfuran pyrolysis. With lower mole fractions of C3 and C4 species in pyrolysis, EF2 had a tendency to produce less aromatic soot precursors than 2-methylfuran and 2,5-dimethylfuran.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work is supported by the National Natural Science Foundation of China (No. 51776045) and the Project Sponsored by the Scientific Research Foundation of Guangxi University (Grant no. XGZ170074).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2020.12.029.

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